



NAME : SHABRINA QOTTRUNNADA

CLASS : SIB 2G

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DATA WAREHOUSE JOBSHEET 2 (OPERATION DATABASE)

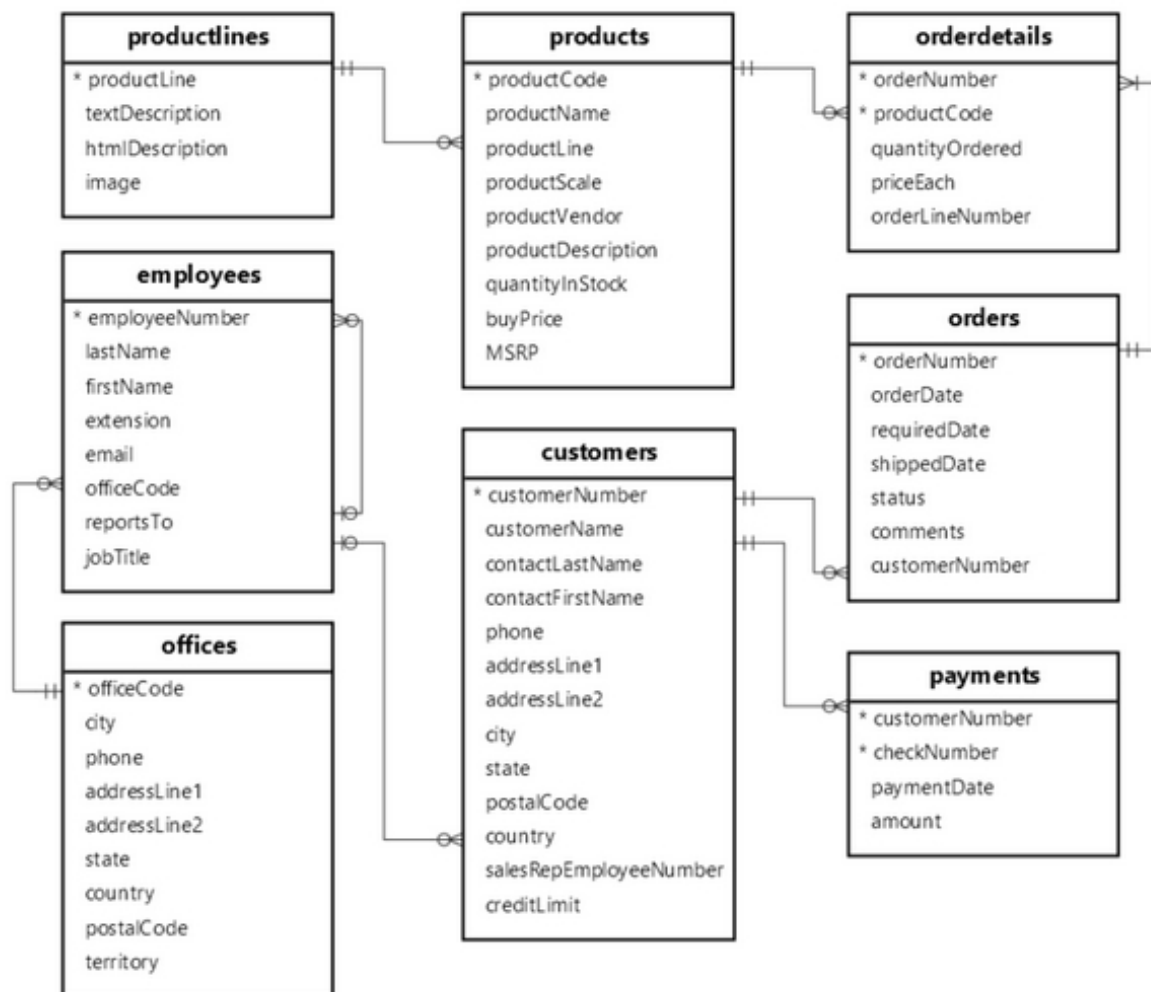
JOBSHEET 2 : DATABASE OPERATIONAL

Tujuan Praktikum

Setelah melakukan praktikum ini, mahasiswa diharapkan dapat lebih mengenal data sumber, cara menganalisa serta melihat kebutuhan baik fungsional maupun non-fungsional dalam pengembangan data warehouse serta lebih memahami apa itu OLTP.

Studi Kasus

LegendVehicle merupakan perusahaan jual-beli tukar-tambah kendaraan klasik. Perusahaan ini memiliki cabang di berbagai negara. LegendVehicle memiliki sistem informasi ERP sendiri. Salah satu modul dari sistem ERP tersebut adalah modul penjualan. Desain database dari modul tersebut adalah sebagai berikut:



Gambar desain modul penjualan

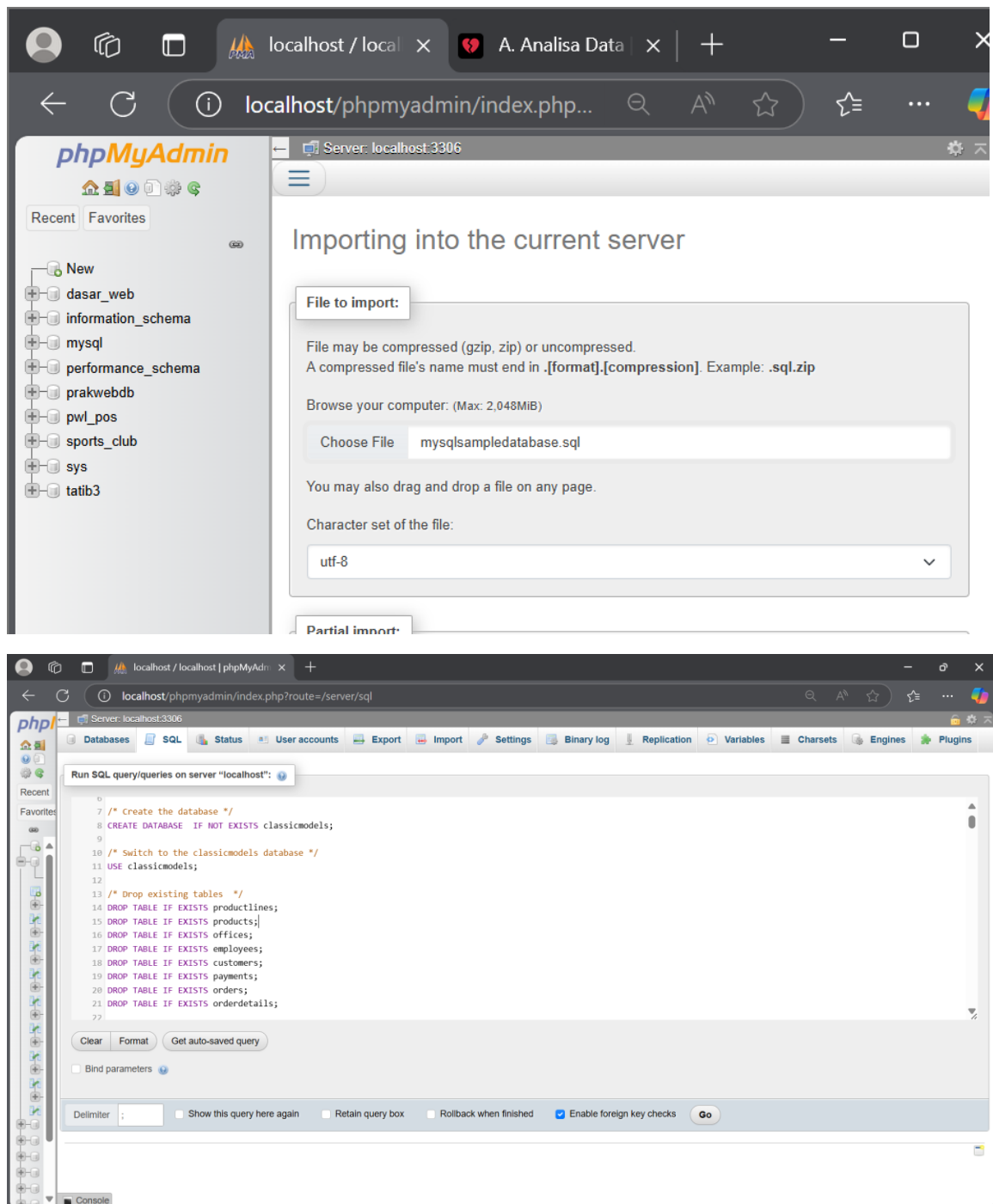
Selain itu proses penjualan kendaraan pada perusahaan tersebut bukan hanya melalui showroom cabang, melainkan reseller-reseller bebas lainnya.

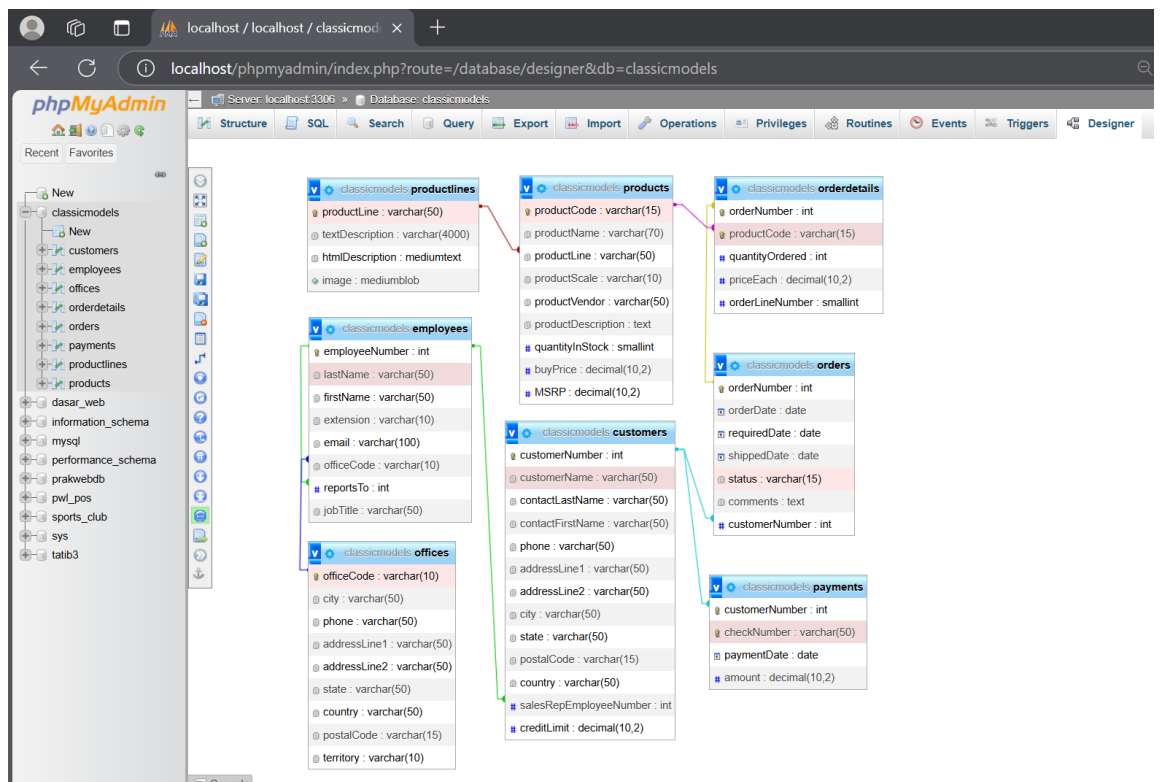
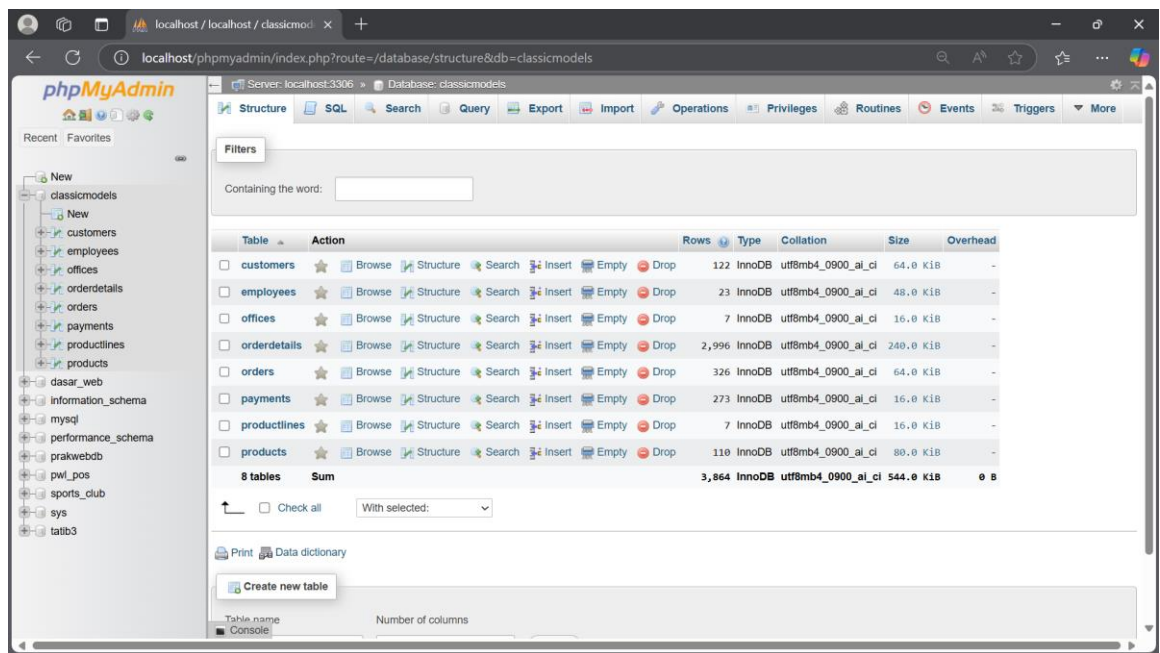
Data penjualan dari cabang tersebut dapat diunduh melalui link berikut: [>> Data Penjualan <<](#)

Tugas 1

1. Import data perusahaan tersebut pada DBMS MySQL!

My Answer :





- Analisa struktur data dari database perusahaan tersebut, dalam bentuk tabel, analisa hubungan setiap tabel nya!

My Answer :

Tabel 1	Tabel 2	Jenis Relasi
productlines	products	One to Many
product	orderdetails	One to Many
customers	orders	One to Many

orders	orderdetails	One to Many
employees	customers	One to Many
employees	employess	One to Many (self join, dikarenakan terdapat reportsTo)
offices	employees	One to Many
customers	payments	One to Many

3. Analisa jumlah field pada setiap tabel!

My Answer :

Nama Tabel	Jumlah Field
productlines	4
products	9
orderdetails	5
orders	6
customers	10
payments	4
employees	7
offices	9

A. ANALISA DATA

PRAKTIKUM 1

1. Jalankan query berikut pada DBMS MySQL yang telah tersedia data Perusahaan LegendVehicle.

```

1 SELECT *
2 FROM employees employee, employees manager, customer cust
3 WHERE employee.reportsTo=manager.employeeNumber
4 AND employee.employeeNumber=cust.salesRepEmployeeNumber;

```

My Answer :

The screenshot shows the phpMyAdmin interface for a database named 'classicmodels'. The SQL query entered is:

```

1 SELECT *
2 FROM employees employee, employees manager, customer cust
3 WHERE employee.reportsTo=manager.employeeNumber
4 AND employee.employeeNumber=cust.salesRepEmployeeNumber;

```

The query is highlighted with a red box. Below the query, there is an error message:

```

Error
SQL query: 1054
SELECT *
FROM employees employee, employees manager, customer cust
WHERE employee.reportsTo=manager.employeeNumber
AND employee.employeeNumber=cust.salesRepEmployeeNumber LIMIT 0, 25

MySQL said:
#1146 - Table 'classicmodels.employees' doesn't exist

```

Hasil dari run kodingan itu error dikarenakan terdapat kesalahan penamaan dalam kodingannya, berikut pemberanannya sesuai dengan tabel yang sudah tersedia.

The top screenshot shows the phpMyAdmin interface with a SQL query being entered in the 'Run SQL query/queries on database classicmodels:' field. The query is:

```
1 SELECT *
2 FROM employees employee, employees manager, customers cust
3 WHERE employee.reportsTo = manager.employeeNumber
4 AND employee.employeeNumber = cust.salesRepEmployeeNumber;
5
```

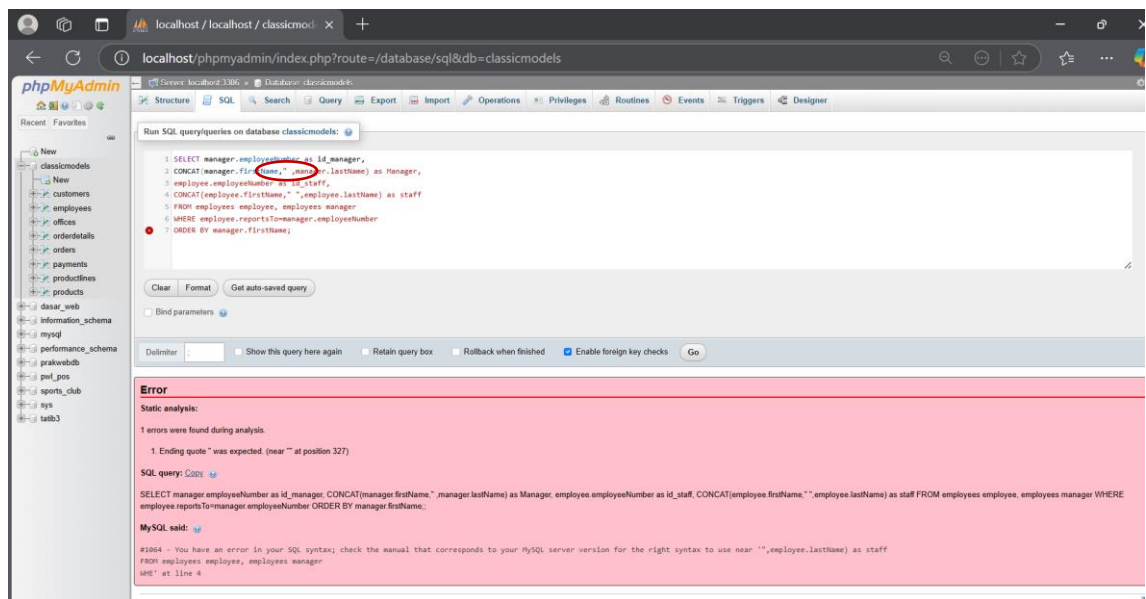
The bottom screenshot shows the result of the query as a table with columns: employeeNumber, lastName, firstName, extension, email, officeCode, reportsTo, jobTitle, employeeNumber, lastName, firstName, extension, email, officeCode, reportsTo, customerNumber, customerName, contactLastName, contactFirstName, phone, addressLine1, addressLine2. The table contains 10 rows of data.

maka hasil dari query tersebut adalah data **Employee** beserta **Manajernya** dan **Customer** yang ia miliki. perhatikan hasil data dengan seksama.

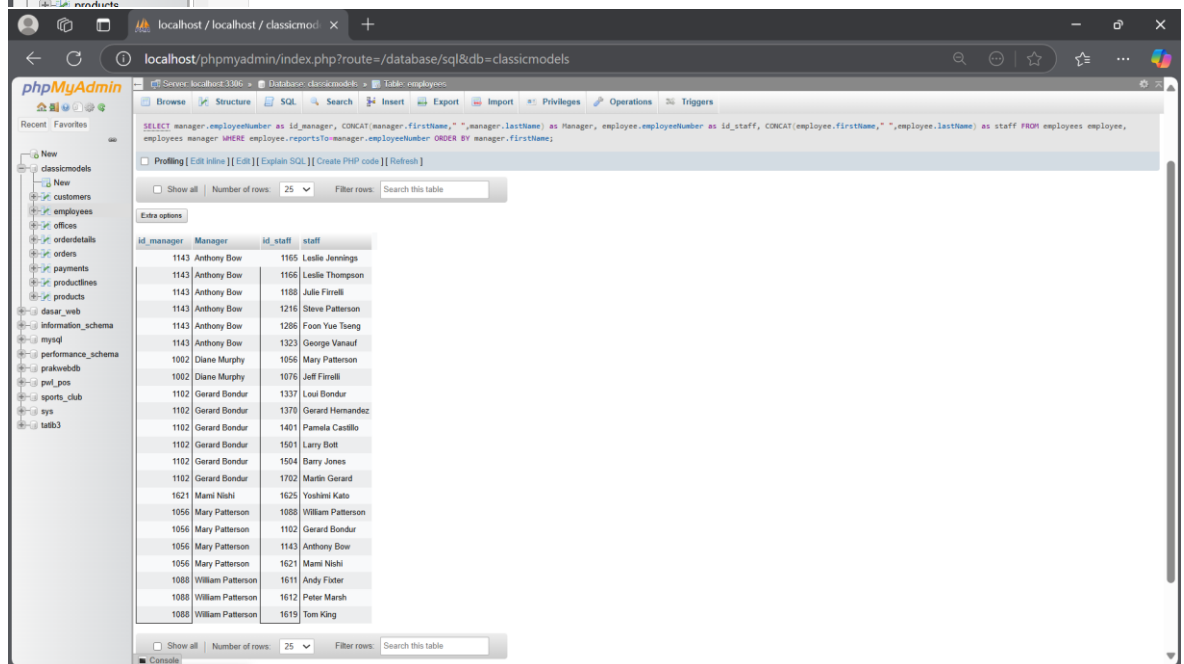
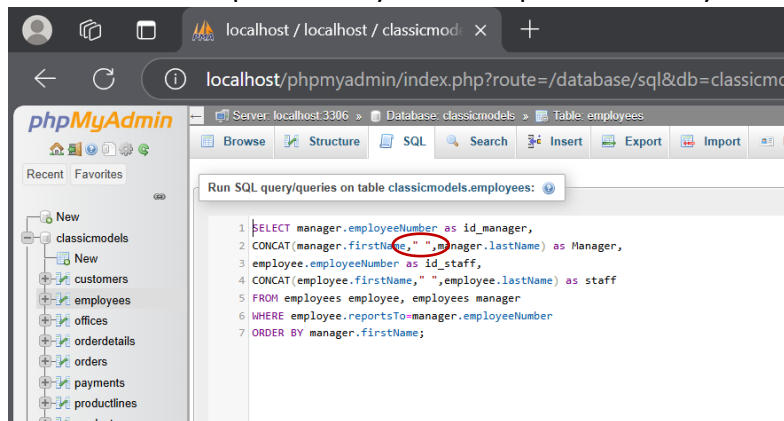
2. Buka **tab baru** pada browser untuk melakukan eksekusi **query** berikut:

```
1 SELECT manager.employeeNumber as id_manager,
2 CONCAT(manager.firstName," ",manager.lastName) as Manager,
3 employee.employeeNumber as id_staff,
4 CONCAT(employee.firstName," ",employee.lastName) as staff
5 FROM employees employee, employees manager
6 WHERE employee.reportsTo=manager.employeeNumber
7 ORDER BY manager.firstName;
```

My Answer :



Setelah saya run di phpMyAdmin (Database) error hasilnya. Dikarenakan terdapat kesalahan dalam penulisannya. Berikut pembenarannya.

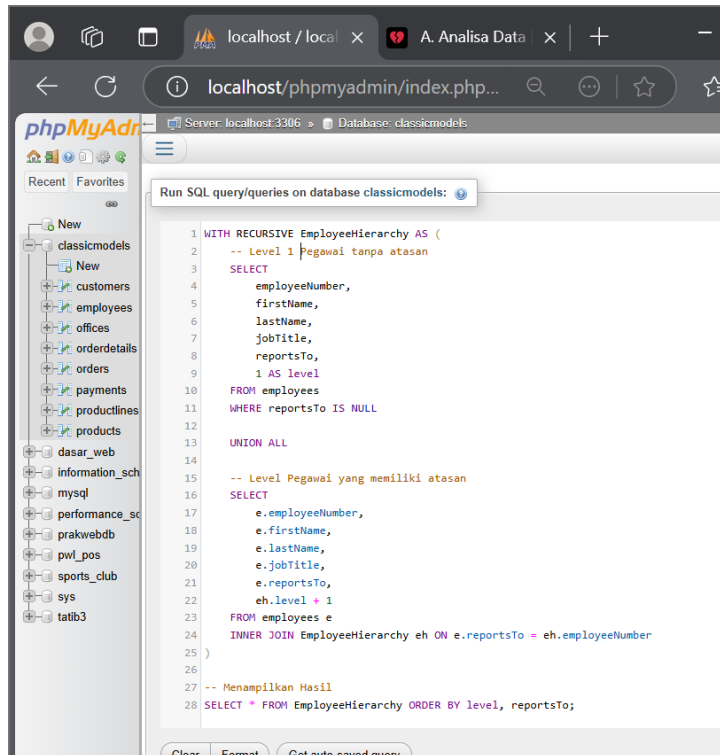


dari hasil query diatas maka akan ditemukan atasan dari setiap pegawai.

TUGAS 2

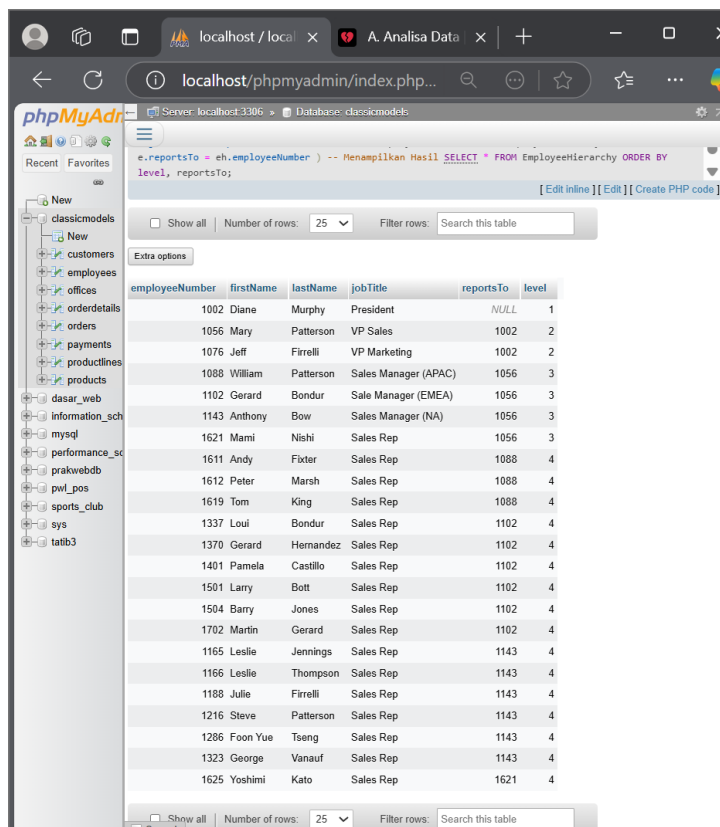
1. Gambarlah hirarki organisasi berdasarkan atasan dari setiap pegawai sesuai dengan hasil praktikum diatas!

My Answer :



The screenshot shows the phpMyAdmin interface with a SQL query editor. The query is a recursive query using a Common Table Expression (CTE) named 'EmployeeHierarchy' to generate an organizational chart. The query is as follows:

```
1 WITH RECURSIVE EmployeeHierarchy AS (  
2   -- Level 1 | pegawai tanpa atasan  
3   SELECT  
4     employeeNumber,  
5     firstName,  
6     lastName,  
7     jobTitle,  
8     reportsTo,  
9     1 AS level  
10  FROM employees  
11  WHERE reportsTo IS NULL  
12  
13  UNION ALL  
14  
15  -- Level Pegawai yang memiliki atasan  
16  SELECT  
17    e.employeeNumber,  
18    e.firstName,  
19    e.lastName,  
20    e.jobTitle,  
21    e.reportsTo,  
22    eh.level + 1  
23  FROM employees e  
24    INNER JOIN EmployeeHierarchy eh ON e.reportsTo = eh.employeeNumber  
25 )  
26  
27 -- Menampilkan Hasil  
28 SELECT * FROM EmployeeHierarchy ORDER BY level, reportsTo;
```



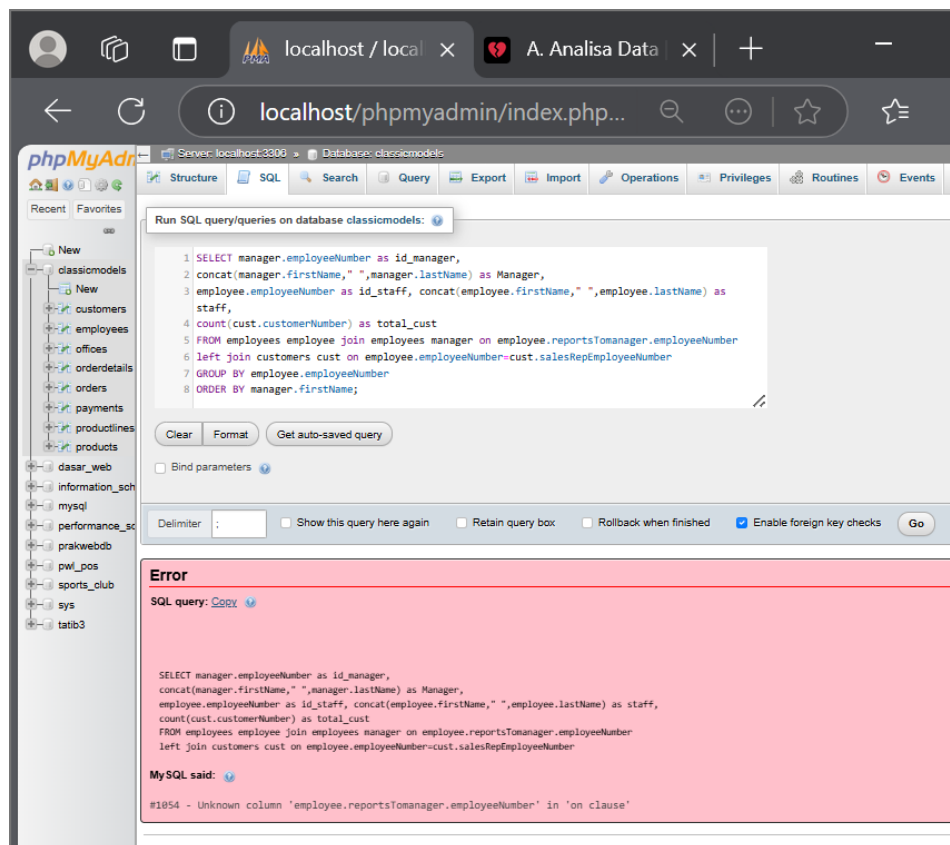
The screenshot shows the results of the recursive SQL query in the phpMyAdmin interface. The results are displayed as a table with the following columns: employeeNumber, firstName, lastName, jobTitle, reportsTo, and level. The table shows the hierarchy of employees, starting from the President (Diane Murphy) at level 1, down to various Sales Representatives at level 4.

employeeNumber	firstName	lastName	jobTitle	reportsTo	level
1002	Diane	Murphy	President	NULL	1
1056	Mary	Patterson	VP Sales	1002	2
1076	Jeff	Firrelli	VP Marketing	1002	2
1088	William	Patterson	Sales Manager (APAC)	1056	3
1102	Gerard	Bondur	Sale Manager (EMEA)	1056	3
1143	Anthony	Bow	Sales Manager (NA)	1056	3
1621	Mami	Nishi	Sales Rep	1056	3
1611	Andy	Fixter	Sales Rep	1088	4
1612	Peter	Marsh	Sales Rep	1088	4
1619	Tom	King	Sales Rep	1088	4
1337	Loui	Bondur	Sales Rep	1102	4
1370	Gerard	Hernandez	Sales Rep	1102	4
1401	Pamela	Castillo	Sales Rep	1102	4
1501	Larry	Bott	Sales Rep	1102	4
1504	Barry	Jones	Sales Rep	1102	4
1702	Martin	Gerard	Sales Rep	1102	4
1165	Leslie	Jennings	Sales Rep	1143	4
1166	Leslie	Thompson	Sales Rep	1143	4
1188	Julie	Firrelli	Sales Rep	1143	4
1216	Steve	Patterson	Sales Rep	1143	4
1286	Foon Yue	Tseng	Sales Rep	1143	4
1323	George	Vanauf	Sales Rep	1143	4
1625	Yoshimi	Kato	Sales Rep	1621	4

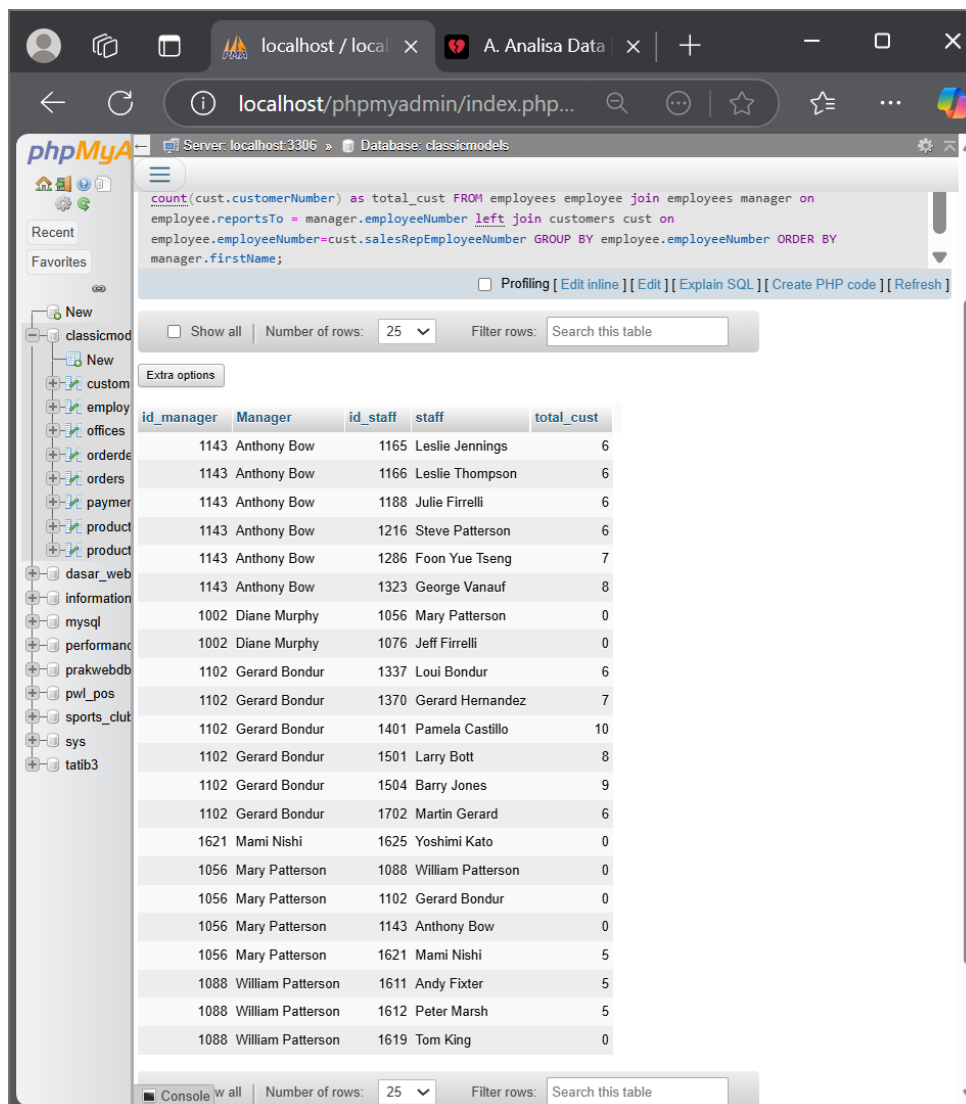
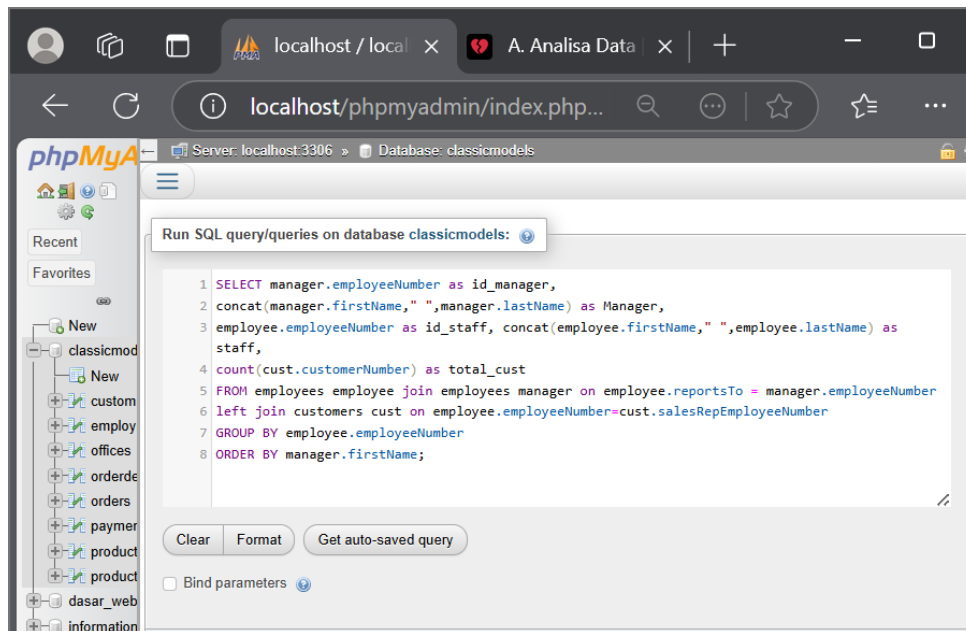
2. Buka **tab baru** pada browser untuk melakukan eksekusi **query** berikut:

```
1 SELECT manager.employeeNumber as id_manager,  
2 concat(manager.firstName," ",manager.lastName) as Manager,  
3 employee.employeeNumber as id_staff, concat(employee.firstName,"  
  ",employee.lastName) as staff,  
4 count(cust.customerNumber) as total_cust  
5 FROM employees employee join employees manager on  
  employee.reportsTomanager.employeeNumber  
6 left join customers cust on  
  employee.employeeNumber=cust.salesRepEmployeeNumber  
7 GROUP BY employee.employeeNumber  
8 ORDER BY manager.firstName;
```

My Answer :



Terdapat error Ketika saya menjalankan query tersebut, hal ini dikarenakan terdapat alias tabel yang tidak ada di tabel yaitu “employees manager on employee.reportsTomanager.employeeNumber”. Seharusnya “employee.reportsTo = manager.employeeNumber” saja dan terdapat juga kesalahan dalam JOIN antar tabel employees. Berikut pembenarannya.



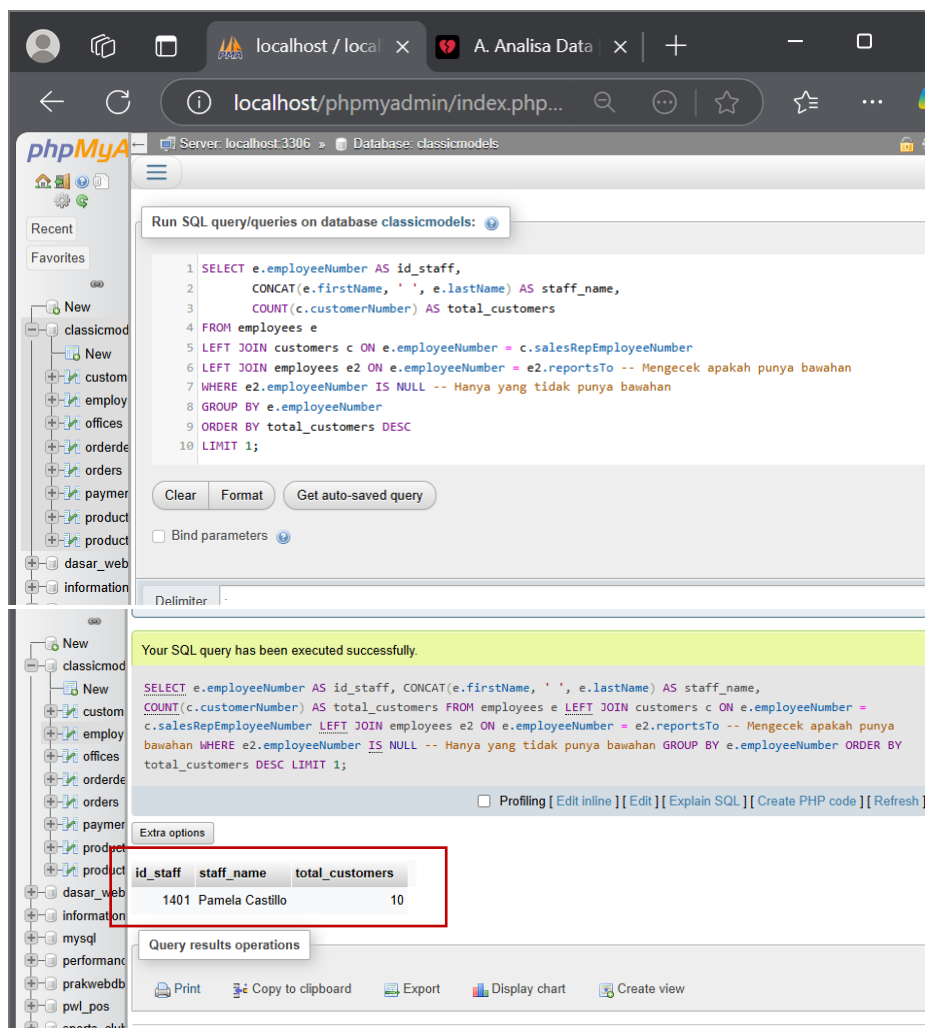
dari query tersebut menghasilkan jumlah **customer** dari setiap **staff**.

Jika perusahaan tersebut memiliki **KPI (Key Performances Indicator) "Jumlah customer yang bertransaksi"** maka jawablah pertanyaan-pertanyaan berikut!

TUGAS 3

1. Siapakah staff dengan hirarki paling bawah yang berprestasi dilihat dari jumlah customer terbanyak?

My Answer :



The screenshot shows the phpMyAdmin interface for a database named 'classicmodels'. The SQL query entered is:

```
1 SELECT e.employeeNumber AS id_staff,  
2       CONCAT(e.firstName, ' ', e.lastName) AS staff_name,  
3       COUNT(c.customerNumber) AS total_customers  
4 FROM employees e  
5 LEFT JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber  
6 LEFT JOIN employees e2 ON e.employeeNumber = e2.reportsTo -- Mengecek apakah punya bawahan  
7 WHERE e2.employeeNumber IS NULL -- Hanya yang tidak punya bawahan  
8 GROUP BY e.employeeNumber  
9 ORDER BY total_customers DESC  
10 LIMIT 1;
```

The query has been executed successfully, and the results are displayed in a table:

id_staff	staff_name	total_customers
1401	Pamela Castillo	10

2. Jika KPI atasan dihitung dari customer yang dimilikinya dijumlah dengan customer dari staff dibawahnya, urutkan ranking prestasi keseluruhan pegawai beserta keterangan jumlah customer yang dimilikinya!

My Answer :

localhost / local x A. Analisa Data x

localhost/phpmyadmin/index.php... Server: localhost:3306 » Database: classicmodels

Run SQL query/queries on database classicmodels:

```
1 WITH CustomerCount AS (  
2   SELECT e.employeeNumber,  
3         e.reportsTo,  
4         COUNT(c.customerNumber) AS direct_customers  
5   FROM employees e  
6  LEFT JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber  
7  GROUP BY e.employeeNumber  
8 ),  
9 KPI AS (  
10  SELECT e1.employeeNumber,  
11        e1.reportsTo,  
12        e1.direct_customers + IFNULL(SUM(e2.direct_customers), 0) AS total_customers  
13  FROM CustomerCount e1  
14  LEFT JOIN CustomerCount e2 ON e1.employeeNumber = e2.reportsTo  
15  GROUP BY e1.employeeNumber  
16 )  
17 SELECT e.employeeNumber AS id_staff,  
18        CONCAT(e.firstName, ' ', e.lastName) AS staff_name,  
19        KPI.total_customers  
20 FROM KPI  
21 JOIN employees e ON KPI.employeeNumber = e.employeeNumber  
22 ORDER BY total_customers DESC;
```

Clear Format Get auto-saved query

☐ Bind parameters

localhost / local x A. Analisa Data x

localhost/phpmyadmin/index.php... Server: localhost:3306 » Database: classicmodels

GROUP BY e.employeeNumber), KPI AS (SELECT e1.employeeNumber, e1.reportsTo, e1.direct_customers + IFNULL(SUM(e2.direct_customers), 0) AS total_customers FROM CustomerCount e1 LEFT JOIN CustomerCount e2 ON e1.employeeNumber = e2.reportsTo GROUP BY e1.employeeNumber) SELECT e.employeeNumber AS id_staff, CONCAT(e.firstName, ' ', e.lastName) AS staff_name, KPI.total_customers FROM KPI JOIN employees e ON

[Edit inline] [Edit] [Create PHP code]

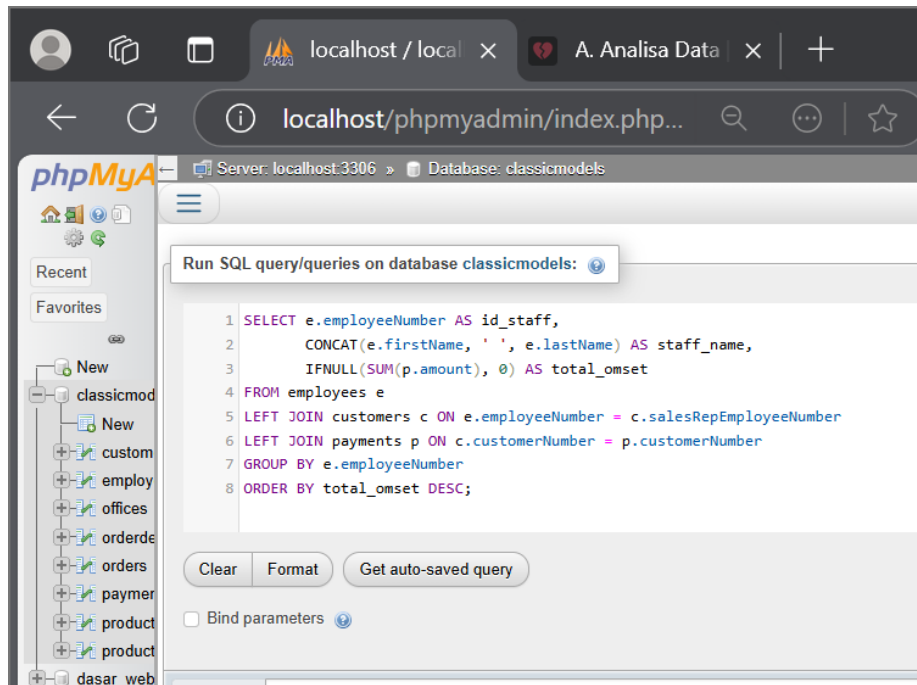
☐ Show all Number of rows: 25 Filter rows: Search this table

Extra options

id_staff	staff_name	total_customers
1102	Gerard Bondur	46
1143	Anthony Bow	39
1088	William Patterson	10
1401	Pamela Castillo	10
1504	Barry Jones	9
1323	George Vanauf	8
1501	Larry Bott	8
1286	Foon Yue Tseng	7
1370	Gerard Hernandez	7
1165	Leslie Jennings	6
1166	Leslie Thompson	6
1188	Julie Firrelli	6
1216	Steve Patterson	6
1337	Loui Bondur	6
1702	Martin Gerard	6
1056	Mary Patterson	5
1611	Andy Fixter	5
1612	Peter Marsh	5
1621	Mami Nishi	5
1002	Diane Murphy	0
1076	Jeff Firrelli	0
1619	Tom King	0
1625	Yoshimi Kato	0

3. Analisa kembali data LegendVehicle untuk mendapatkan ranking pegawai berdasarkan KPI "Jumlah omset yang didapat". Urutkan ranking pegawai beserta keterangan dana yang didapat!

My Answer :



Showing rows 0 - 22 (23 total, Query took 0.0026 seconds.)

SELECT e.employeeNumber AS id_staff, CONCAT(e.firstName, ' ', e.lastName) AS staff_name, IFNULL(SUM(p.amount), 0) AS total_omset FROM employees e LEFT JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber LEFT JOIN payments p ON c.customerNumber = p.customerNumber GROUP BY e.employeeNumber ORDER BY total_omset DESC;

Extra options

id_staff	staff_name	total_omset
1370	Gerard Hernandez	1112003.81
1165	Leslie Jennings	989906.55
1401	Pamela Castillo	750201.87
1501	Larry Bott	689653.25
1504	Barry Jones	637672.65
1323	George Vanauf	584406.80
1337	Loui Bondur	569485.75
1611	Andy Fixter	509385.82
1612	Peter Marsh	497907.16
1286	Foon Yue Tseng	488212.67
1621	Mami Nishi	457110.07
1216	Steve Patterson	449219.13
1702	Martin Gerard	387477.47
1188	Julie Firrelli	386663.20
1166	Leslie Thompson	347533.03
1002	Diane Murphy	0.00
1056	Mary Patterson	0.00
1076	Jeff Firrelli	0.00
1088	William Patterson	0.00
1102	Gerard Bondur	0.00
1143	Anthony Bow	0.00
1619	Tom King	0.00
1625	Yoshimi Kato	0.00

4. Jika KPI yang pertama merupakan "Jumlah customer yang bertransaksi" sedangkan KPI yang kedua "Jumlah omset yang didapat". Maka, berapakah jumlah field yang dibutuhkan untuk mendapatkan informasi tersebut?

My Answer :

KPI	Jumlah Field
Jumlah customer yang bertransaksi	1
Jumlah omset yang didapatkan	1

Penjelasan

- Jumlah customer yang bertransaksi → bisa kita dapatkan dengan jumlah unik customerNumber dalam tabel payments, karena hanya pelanggan yang melakukan transaksi yang dihitung.
- Jumlah omset yang didapat → bisa kita dapatkan dengan menjumlahkan nilai amount dari tabel payments, karena field menyimpan total pembayaran dari setiap transaksi pelanggan.

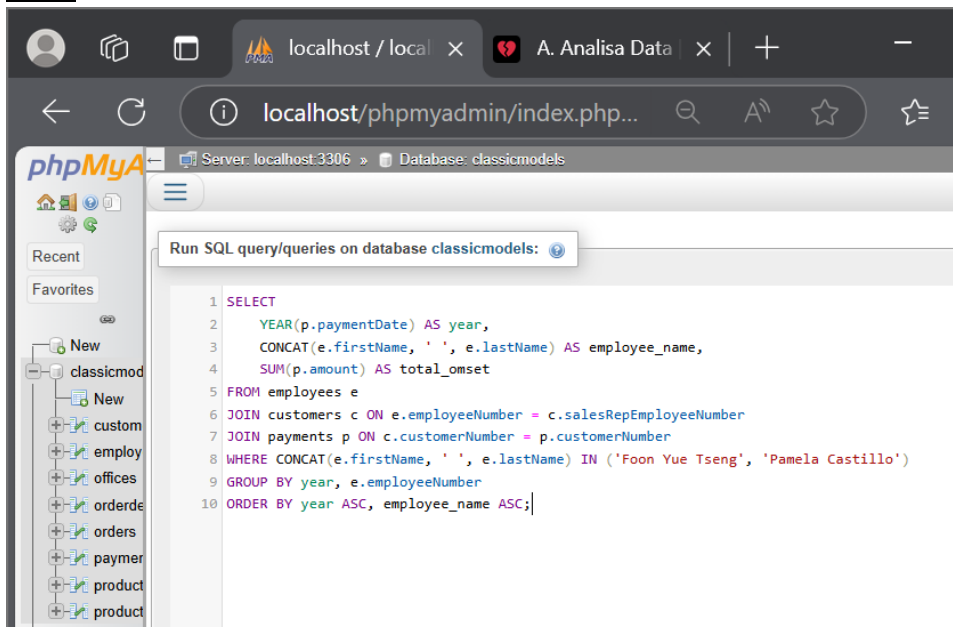
5. Buatlah report pertahun untuk KPI "Jumlah omset yang didapat" pada Foon Yue Tseng dan Pamela Castillo. Serta gambarkan grafiknya (grafik garis).

Nama	2003	2004	2005
Foon Yue Tseng			
Pamela Castillo			

My Answer :

Query

Input



The screenshot shows the phpMyAdmin interface with the following SQL query entered in the query editor:

```

1 SELECT
2     YEAR(p.paymentDate) AS year,
3     CONCAT(e.firstName, ' ', e.lastName) AS employee_name,
4     SUM(p.amount) AS total_omset
5 FROM employees e
6 JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
7 JOIN payments p ON c.customerNumber = p.customerNumber
8 WHERE CONCAT(e.firstName, ' ', e.lastName) IN ('Foon Yue Tseng', 'Pamela Castillo')
9 GROUP BY year, e.employeeNumber
10 ORDER BY year ASC, employee_name ASC;

```

Output

Showing rows 0 - 5 (6 total, Query took 0.0016 seconds.)

```
SELECT YEAR(p.paymentDate) AS year, CONCAT(e.firstName, ' ', e.lastName) AS employee_name, SUM(p.amount) AS total_omset FROM employees e JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber JOIN payments p ON c.customerNumber = p.customerNumber WHERE CONCAT(e.firstName, ' ', e.lastName) IN ('Foon Yue Tseng', 'Pamela Castillo') GROUP BY year, e.employeeNumber ORDER BY year ASC, employee_name ASC;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

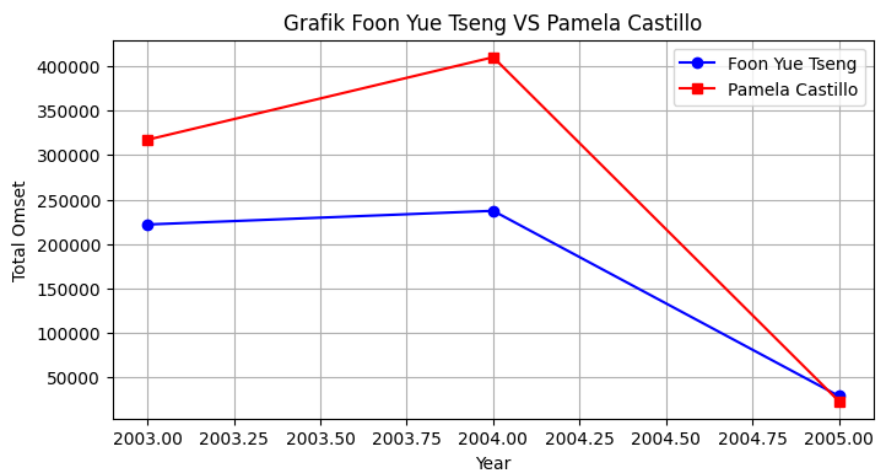
year	employee_name	total_omset
2003	Foon Yue Tseng	221887.03
2003	Pamela Castillo	317104.78
2004	Foon Yue Tseng	237255.26
2004	Pamela Castillo	409910.07
2005	Foon Yue Tseng	29070.38
2005	Pamela Castillo	23187.02

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

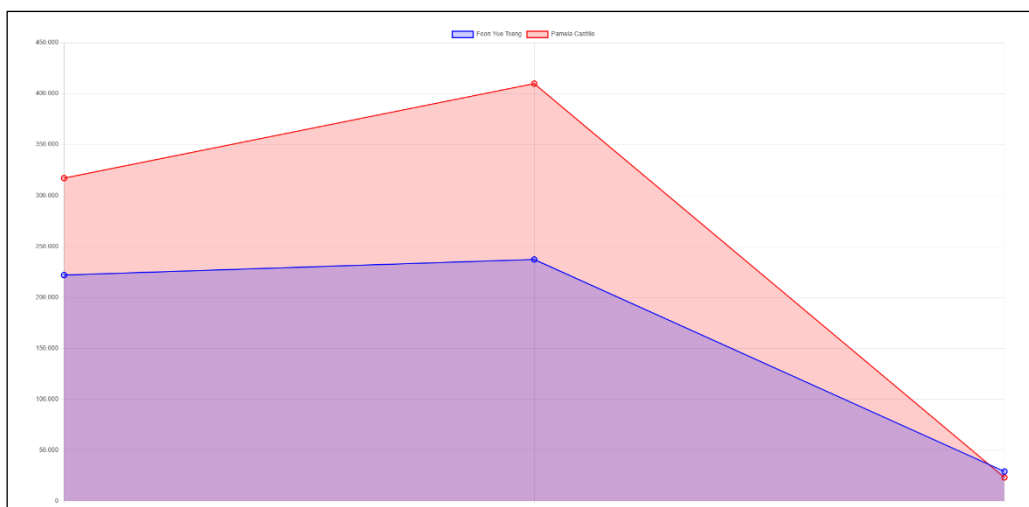
chart

From Google Colab



From php

Grafik Foon Yue Tseng vs Pamela Castillo



STUDI KASUS

Pak Huhut merupakan pemegang saham LegendVehicle. dia membutuhkan dashboard untuk melihat perkembangan penjualan (omset) di setiap cabang di tiap tahunnya. Dikarenakan perusahaan tersebut belum merekrut Data Engineer maka, penarikan informasi hanya bisa dilakukan melalui OLTP yang ada.

Hasil report yang diinginkan adalah grafik berdasarkan tabel berikut:

Nama Cabang	2003	2004	2005
...			
...			

Analisalah terlebih dahulu:

1. Field apa saja yang diperlukan untuk menampilkan penjualan di setiap cabang.

My Answer :

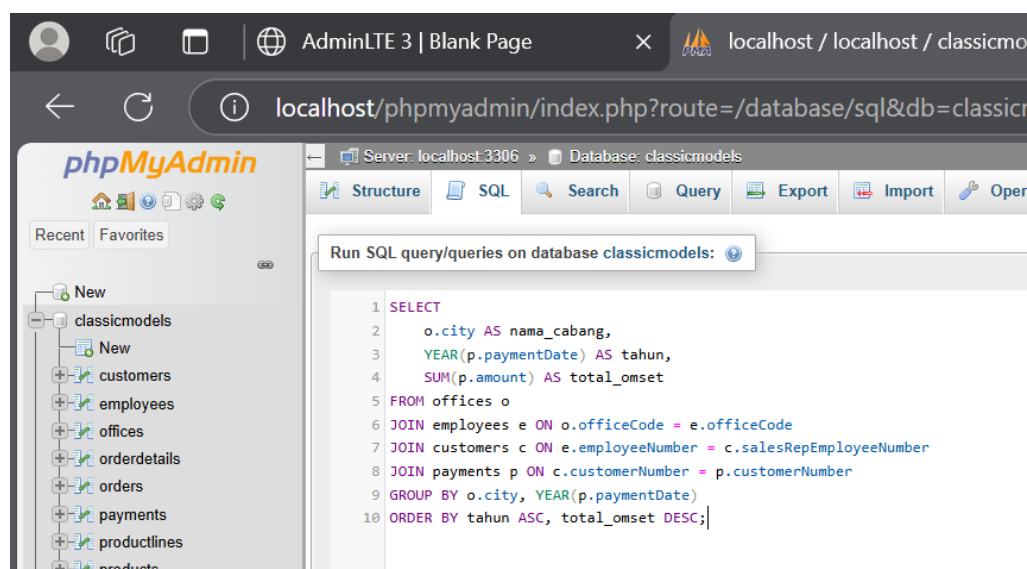
- Nama Cabang : diambil dari tabel offices, (offices.officesName)
- Tahun Transaksi : diambil dari tabel payment, YEAR(payments.paymentDate) AS Tahun
- Total Omset : diambil dari total pembayaran customer selama setahun, SUM(payments.amount) AS total_omset

2. Bentuk query dengan memperhatikan relasi antar tabel.

My Answer :

Query

Input



Output

phpMyAdmin

Server: localhost:3306 » Database: classicmodels » Table: offices

Showing rows 0 - 20 (21 total, Query took 0.0039 seconds.)

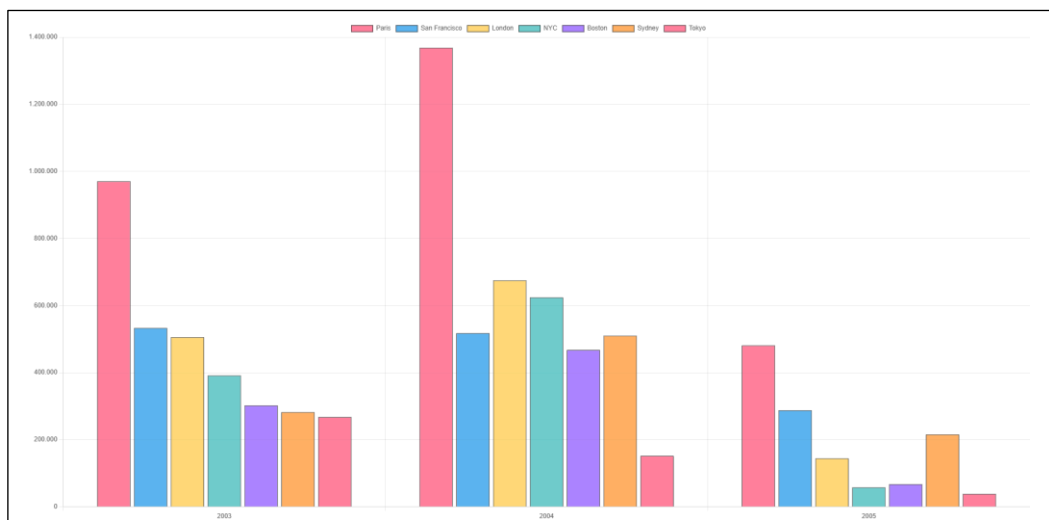
```
SELECT o.city AS nama_cabang, YEAR(p.paymentDate) AS tahun, SUM(p.amount) AS total_omset
FROM payments p ON c.customerNumber = p.customerNumber GROUP BY o.city, YEAR(p.paymentDate)
```

Extra options

nama_cabang	tahun	total_omset
Paris	2003	969959.90
San Francisco	2003	532681.13
London	2003	505384.85
NYC	2003	391175.53
Boston	2003	301781.38
Sydney	2003	281985.51
Tokyo	2003	267249.40
Paris	2004	1368458.96
London	2004	674815.75
NYC	2004	623872.78
San Francisco	2004	517408.62
Sydney	2004	509833.62
Boston	2004	467177.07
Tokyo	2004	151761.45
Paris	2005	480750.04
San Francisco	2005	287349.83
Sydney	2005	215473.85
London	2005	144125.30
Boston	2005	66923.88
NYC	2005	57571.16
Tokyo	2005	38099.22

chart

From php



SOAL BONUS: buatlah report lain dengan sumber data OLTP yang sama, analisa field yang digunakan, bentuk struktur query dan tuliskan dalam tabel serta grafiknya.

My Answer

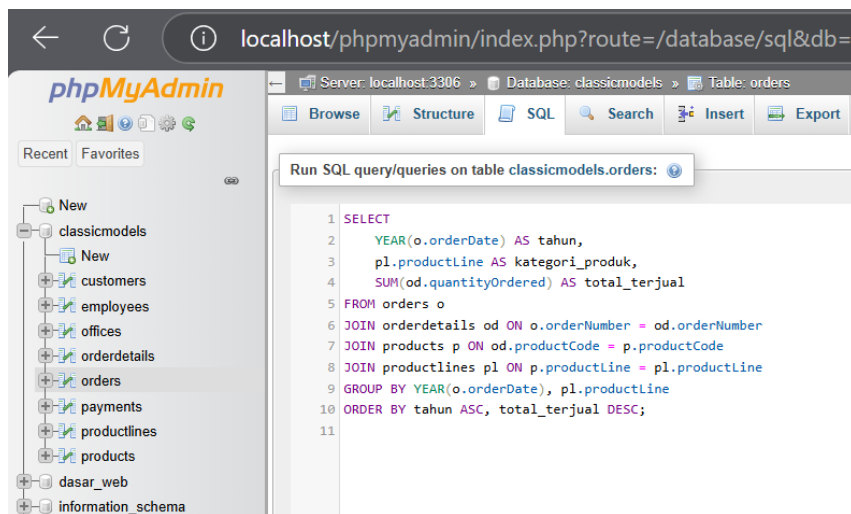
STUDI KASUS : Kategori Produk yang Terlaris Dijual per Tahunnya

Field yang digunakan :

- Tahun Transaksi : diambil dari tabel orders, YEAR(orders.orderDate)
- Kategori Produk : diambil dari tabel productlines, (productlines.productLine)
- Total Terjual : diambil dari tabel orderdetails, SUM(orderdetails.quantityOrdered)

Query :

input



output

The screenshot shows the phpMyAdmin interface displaying the results of the SQL query. The results are shown in a table with 3 columns: tahun, kategori_produk, and total_terjual. The table contains 21 rows of data, showing the total quantity ordered for each product line grouped by year.

tahun	kategori_produk	total_terjual
2003	Classic Cars	12762
2003	Vintage Cars	7913
2003	Trucks and Buses	4056
2003	Motorcycles	4031
2003	Planes	3833
2003	Ships	2844
2003	Trains	1000
2004	Classic Cars	16085
2004	Vintage Cars	10864
2004	Motorcycles	5976
2004	Planes	5820
2004	Trucks and Buses	5024
2004	Ships	4309
2004	Trains	1409
2005	Classic Cars	6735
2005	Vintage Cars	4156
2005	Motorcycles	2771
2005	Planes	2219
2005	Trucks and Buses	1921
2005	Ships	1379
2005	Trains	409

Chart

